

BICONOMY

The Infrastructure Layer for Autonomous Finance

WHITE PAPER

ABSTRACT

The central problem in fintech today is the lack of infrastructure for AI to execute through. AI can process markets and make financial decisions in milliseconds, but the systems that implement those decisions remain people based. Biconomy is a decentralized financial infrastructure layer built on account abstraction and smart sessions, enabling autonomous AI agents to execute, batch, and settle transactions within a set of permissioned environments. Biconomy's core architecture provides a programmable execution layer for AI agents via account abstraction, which converts traditional cryptocurrency wallets into programmable smart contracts. The operational boundaries, spending limits, allowed contracts, and temporal expirations, are hardcoded onto the blockchain to help mitigate the chance that an AI goes rogue, hallucinates a bad trade, or steals money. The market opportunity for Biconomy lies in Blockchain infrastructure, Agentic AI, and Digital payments/Fintech. Web3 users are expected to grow to USD 29.97 billion by 2031. Biconomy's core differentiators are its transaction abstraction, programmable smart account infrastructure, and user-centric automation designed to support seamless Web3 and AI-agent interactions. Although Biconomy offers a compelling solution, it is hindered by significant risks including regulatory uncertainty, smart contract and infrastructure failures, and adoption frictions.

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The Problem

The Infrastructure Gap in Autonomous Finance

Existing fintech infrastructure and rails were not designed to support AI execution. AI can process markets and make financial decisions in milliseconds, but the systems that implement those decisions remain people-based (Ifeanyichukwu and Vaswani, 2025). Banking systems and common blockchain wallets are designed for manual procedures in which users approve individual transactions (Arshadi and Dombrowski, 2026).

AI has become a core enabler in the fintech space, enabling automated analysis and optimized decision-making (Cao et al. 2020). There are three dimensions to this problem. First, decentralized financial systems remain disjointed, and cross-chain transactions often require multiple approvals across multiple wallets and protocols (Gong, 2026). Additionally, there are five to eight confirmations within a single DeFi operation before it is executed (Xue et al., 2023). Second, gas fee structures remain unpredictable, and users must hold native tokens across all blockchain ecosystems, limiting the scalability of automation (Tseng et al., 2026). Third, the most common type of blockchain wallet is an Externally Owned Account (EOA), which, unfortunately, lacks the features necessary for safe AI financial activity (Stoica, 2023).

Currently, AI requires constant human approval for transactions. Blockchain offers trustless settlement, but can't adapt, read intent, or assess risk in real time. They are complementary but lack a unified infrastructure to work together.

Emerging Opportunity

DeFi has grown complex enough that manual management is no longer feasible, and regulators under MiCA and SEC guidance are now demanding the kind of auditable execution that on-chain infrastructure naturally provides. Nonetheless, the need for this shift is further emphasized by broader industry trends. For example, DeFi has become a \$100 billion ecosystem, with the AI-fintech market expected to exceed \$60 billion by 2031, as shown in Figure 1. According to Cobo (2022), cryptocurrency provides the ideal environment for AI agents to thrive; for instance, 24/7 markets that never sleep, such as crypto markets that trade constantly across every time zone, will require automated bots to trade with the same ability as a human. Therefore, it is essential to

address the Blockchain UX gap by adopting the right infrastructure to mitigate the complexity of Web3 applications for AI agents (Vasconcelos, 2026).

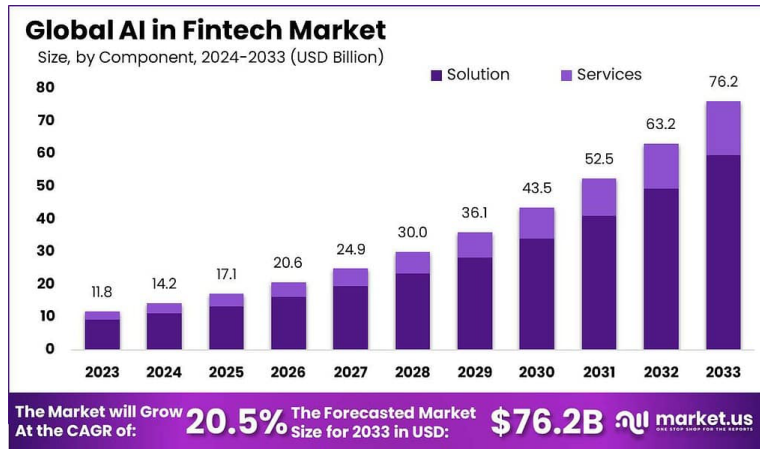


Figure 1. AI-Fintech Market growth from 2023 and forecast to 2033 (Market.us, 2024)

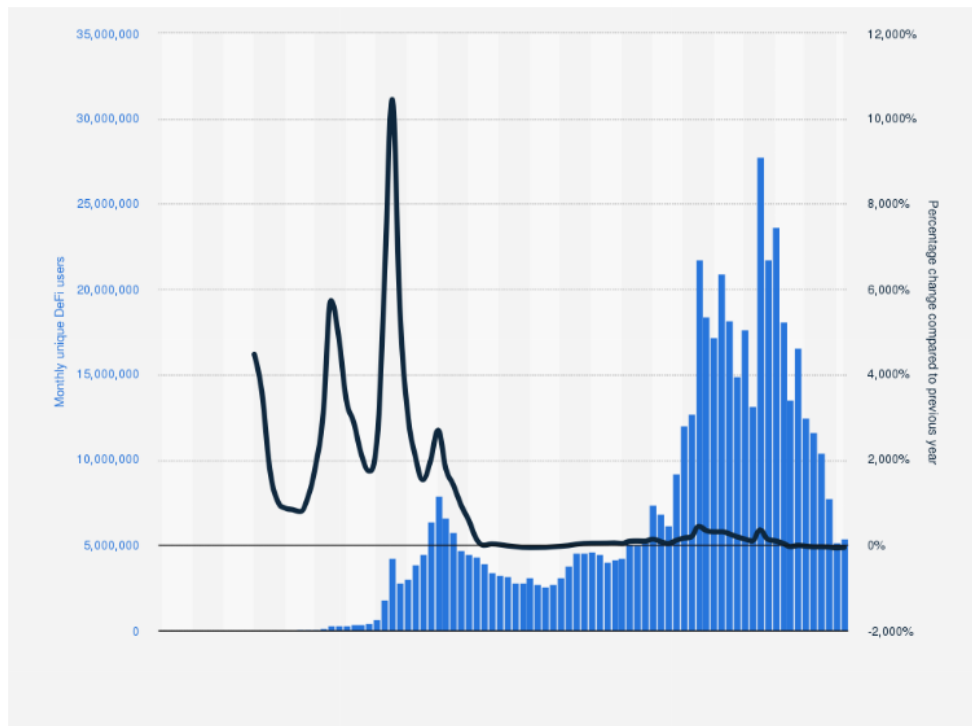


Figure 2. DeFi user accounts from 2018 to 2026 (Scholle, 2026)

The Solution: Biconomy

Agentic Commerce via Account Abstraction

Biconomy is a DeFi infrastructure layer built on account abstraction and smart sessions, enabling autonomous AI agents to execute, batch, and settle transactions within a set of permissioned

environments. Biconomy's core architecture uses the ERC-4337 standard, which converts traditional cryptocurrency wallets into programmable smart contracts (Lin et al., 2024). ERC-4337 has already facilitated over 20 million user operations, as programmable account systems gain traction. Smart Accounts provide a way to abstract validation logic, enabling contract wallets to create transactions independently without requiring an EOA to trigger them (Chin, Emura, & Omote, 2025). This architecture enables users to create Smart Sessions, which are delegated versions of the user's execution keys that provide limited permissions for an AI agent. Moreover, infrastructure services such as Paymasters handle gas fees, enabling the AI to transact without managing native tokens for compute (Kumar Singh et al., 2023).

Divergence from Traditional Fintech

Legacy rails rely on traditional approaches to identity verification, including physical or digital signatures, and on banking intermediaries. Biconomy leverages AI to create a programmatic on-chain execution identity for the AI on the blockchain and, through the programmability and abstraction of accounts, fully separates transaction authorization from simple key possession (Lin et al., 2024). The main advantage of this architecture is trust-minimized automation. Users get the speed, scale, and uninterrupted operation of a stand-alone financial assistant, binding the AI to its instructions. It is bound by spending limits, allowed contracts, and temporal expirations that are hardcoded into the blockchain.

Mechanics of AI Layer

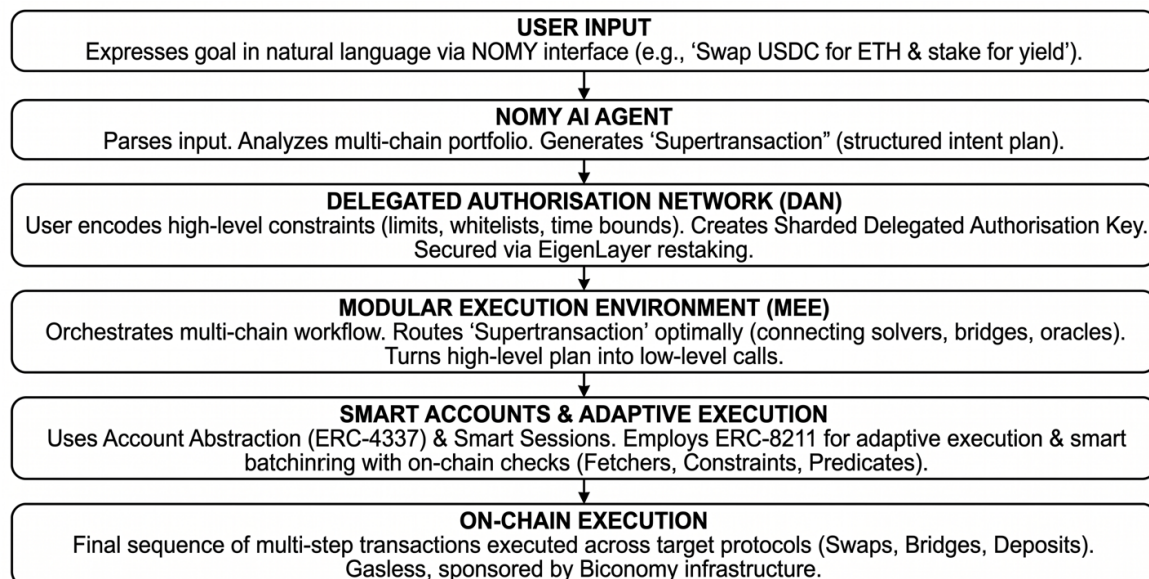


Figure 3: Biconomy AI Workflow: From user intent to execution

NOMY

Intent interface

NOMY is Biconomy's AI agent, to which users chat to convey their goals (Biconomy, 2026a). The AI agent then evaluates this message and creates a responding transaction for the user. NOMY will then aggregate a user's assets across wallets and chains into a unified view, suggesting personalized recommendations for the user's positions (Biconomy, 2026a). The agent decomposes each intent into a sequence of actions, such as swaps, bridges, approvals, and deposits.

Delegated Authorization Network (DAN)

The Delegated Authorization Network layer allows users to grant AI agents specific powers, with permissions enforced by EigenLayer's restaked validators (Biconomy, 2025b). The EigenLayer is a protocol that runs on top of Ethereum, enabling restaking and allowing users to reuse their staked ETH to secure other applications (EigenCloud, 2026). DAN will create a delegated authorization key for every user, which is sharded across nodes. This reduces the need to expose the user's raw private key to the AI (Biconomy, 2025b).

Modular Execution Environment (MEE)

The Modular Execution Environment is an orchestration engine that executes NOMY's AI-generated super transactions on the user's behalf (Biconomy, 2025a). Super transactions are a single signed data model that represents multiple actions and intents of transactions for orchestration across chains. The key innovation here is that it can orchestrate across different chains, coordinating components such as solvers, bridges, oracles, and liquidity sources (Biconomy, 2025a).

Smart Features, Execution & ERC-8211

Biconomy has built on existing smart contracts, enabling programmable wallets with custom validation rules, batched transactions, and sponsored gas (Biconomy, 2026b). Sponsored gas is a feature that allows dApp developers to pay transaction fees for their users (Lim, 2025). Biconomy proposed ERC-8211, a new standard that helps outline how AI agents can safely execute complex DeFi strategies (Mislav, 2026).

Mechanics of the Blockchain layer

Biconomy's blockchain uses an ERC-4337-based account abstraction stack comprising of Smart Accounts, Bundlers, Paymasters, and Smart sessions (Biconomy, 2026a). Smart Accounts have custom validation rules that are checked before any operation goes through (Biconomy, 2025a). The validated logic in each account enforces these policy constraints (Biconomy, 2025a).

With a Smart Session, a user can add a session module to their Smart Account. Giving a separate session key to an agent with certain permissions (Biconomy, 2026b). Each session has its own policies that control which contract addresses, function selectors, parameter rules, value limits, time frames, and networks the key can access (Biconomy, 2025b). When an agent signs a UserOperation using the session key, the Smart Account checks if the operation meets the session policy before sending it to the EntryPoint. If any rule is broken, validation fails, and nothing changes on the blockchain (Biconomy, 2026b).

Biconomy uses Paymasters and Bundlers to handle gas (Biconomy, 2026a). Instead of having users or agents pay for every transaction in ETH, a Paymaster contract can cover gas costs or accept ERC-20 tokens as payment (Biconomy, 2025c). Bundlers collect many UserOperations and send them to the shared EntryPoint contract (Biconomy, 2026a). This setup means agents do not have to manage gas balances on different chains (Biconomy, 2026a). This is especially useful because autonomous agents often make many small transactions, which would be hard to manage manually (Biconomy, 2025d).

ERC-8211 lets agents describe a series of actions as an off-chain script, which is checked against the current on-chain state when it runs (CryptoNews, 2026). Each parameter in a batch includes a fetcher, constraints, and optimal predicate entries (Biconomy and Ethereum Foundation, 2026a). This setup allows agents to execute strategies subject to predefined constraints, such as swapping token A for token B and then depositing the resulting token into a lending protocol, if collateral ratios remain above a set limit (BlockEden, 2026).

STEP-BY-STEP TRANSACTION FLOW

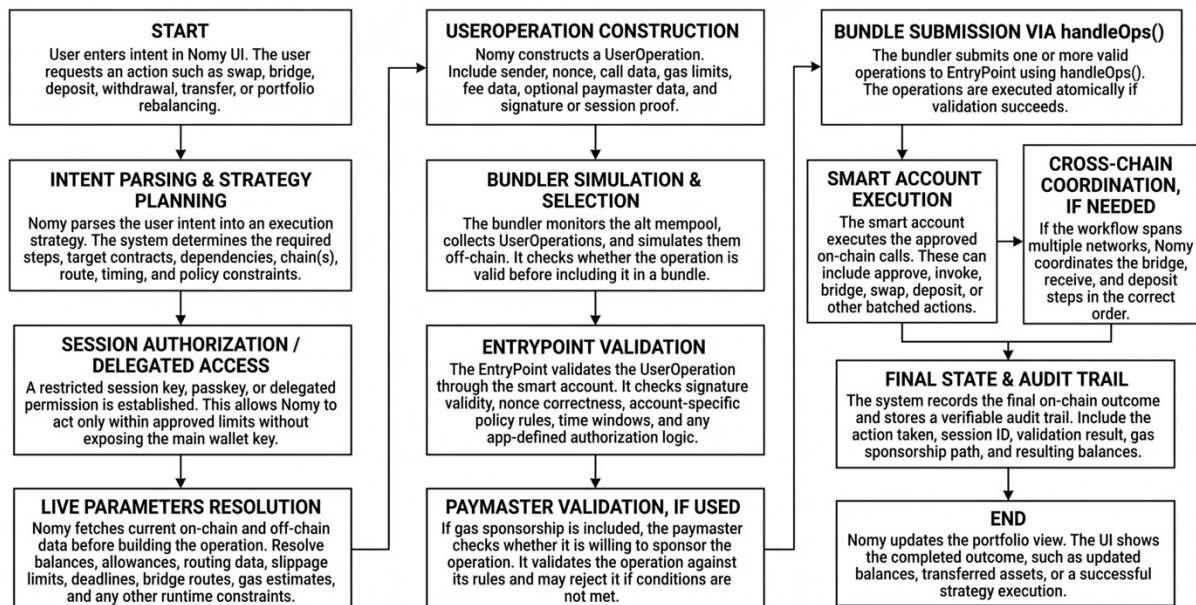


Figure 4: Complete Theoretical Transaction Process for Biconomy

How Biconomy Infrastructure Supports Blockchain

Biconomy is the multichain transaction infrastructure for next-generation Web 3.0 applications, simplifying blockchain transactions for developers to build on and for end-users to interact with (Yash Argarwal, 2022). Biconomy architecture network includes the following features:

- A decentralized network of work of nodes/relayers, which allows users to connect without paying gas via gasless transactions, and purchase gas in their preferred ERC20 or dApp token.
- Eliminates additional blockchain complexity, such as network switching, and allows consumers to benefit from quick transfers.

Three main products of Biconomy are: Mexa, Forward, and Hyphen, each one of them solving challenges like onboarding/UX, gas fees, and interoperability.

1. **Mexa** - this is a JavaScript-based implementation that can be easily integrated into a dApp to support gasless transactions. Biconomy does this through meta transactions, in which Biconomy relayers pay the gas fees, which are then reimbursed to the dApp developer at the end of the month. Examples of dApps using Mexa (Appendix B).

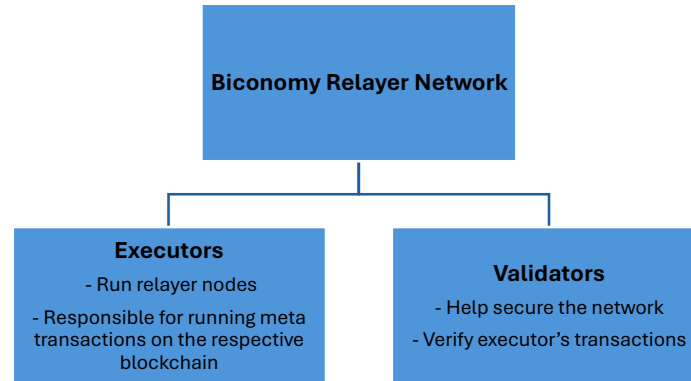


Figure 5: Biconomy Relay Network

- Hyphen - supports interoperability and faster, less expensive token transfers between different blockchains (cross-chain transfers). This ensures instant, cheaper (gas-efficient) transfers whilst maintaining the liquidity of the tokens on both ends of the chain (Appendix B)

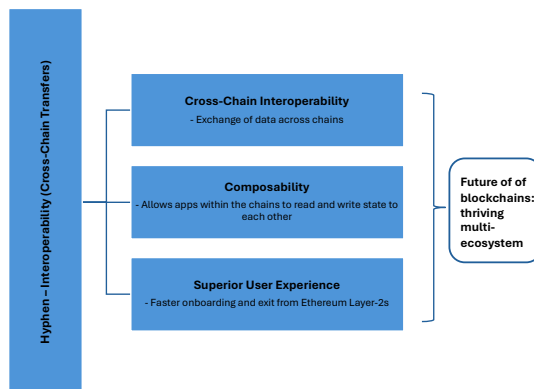


Figure 6: Interoperability, Biconomy

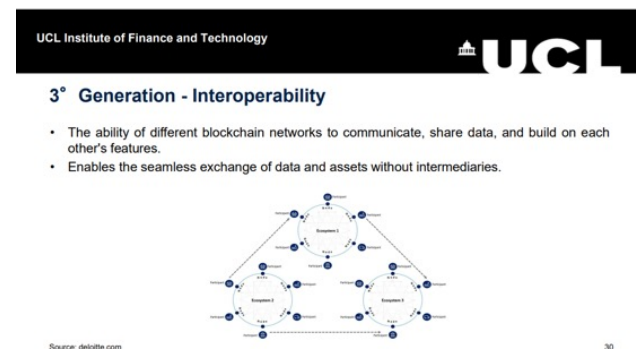


Figure 7: Interoperability Summary, UCL IFTE0014

- Forward - allows users to pay transaction fees with any ERC-20 tokens, a feature not available on the Ethereum blockchain. For dApps like DeFi, this is a huge problem as not all users may have the funds to cover the gas in ETH and with a huge volume of transactions maintenance is a huge challenge (Appendix B). Biconomy solves this challenge ensuring transactions are not stuck as relayers are able to automatically detect congestion and gradually increase gas fees to maintain transaction flow, improving user experience.

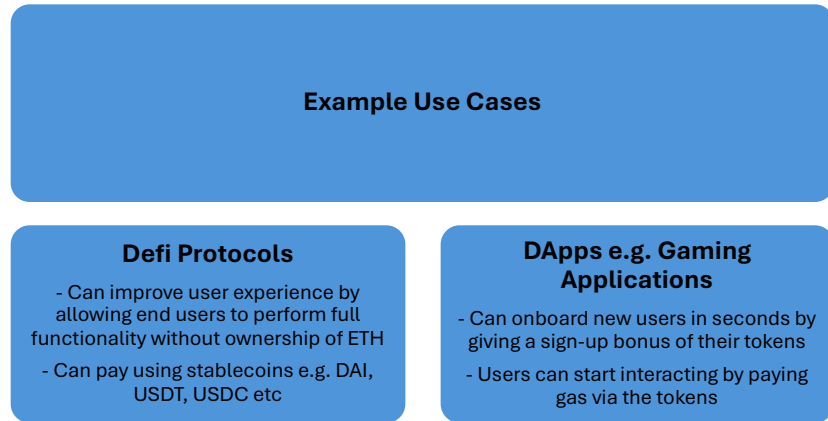


Figure 8: Example Use Cases for Forward (Biconomy Product)

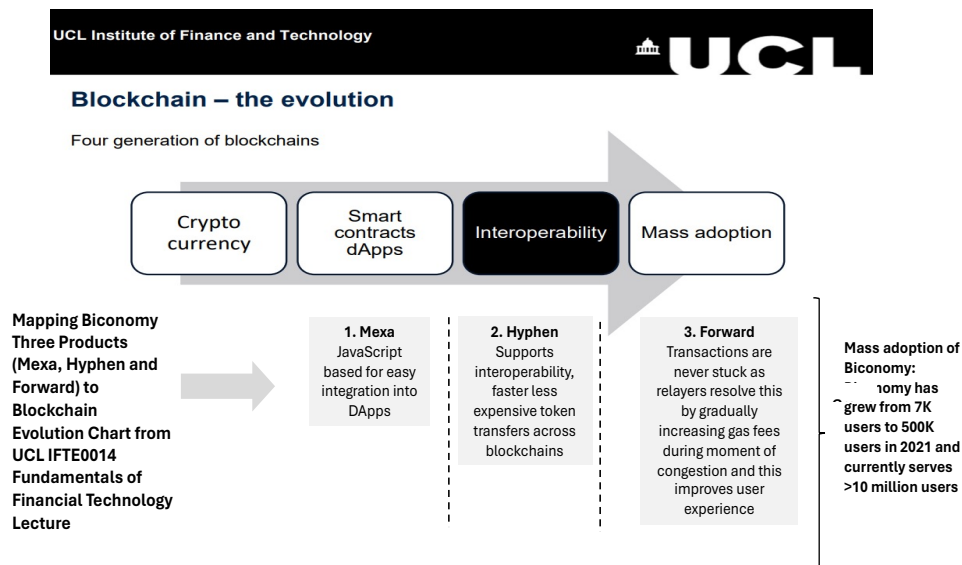


Figure 9: Mapping Biconomy Three Products (Mexa, Hyphen, and Forward) to Blockchain Evolution Chart from UCL IFTE0014 Lecture

Market Size and Opportunity

The market opportunity for Biconomy is in Blockchain infrastructure, Agentic AI, and Digital payments/Fintech, all of which are rapidly growing. The execution layer sits at the intersection of these market needs (Blakey, 2026). The autonomous agentic AI market has grown from \$4.42 billion in 2025 to \$5.83 billion in 2026 (Trading News, 2026).

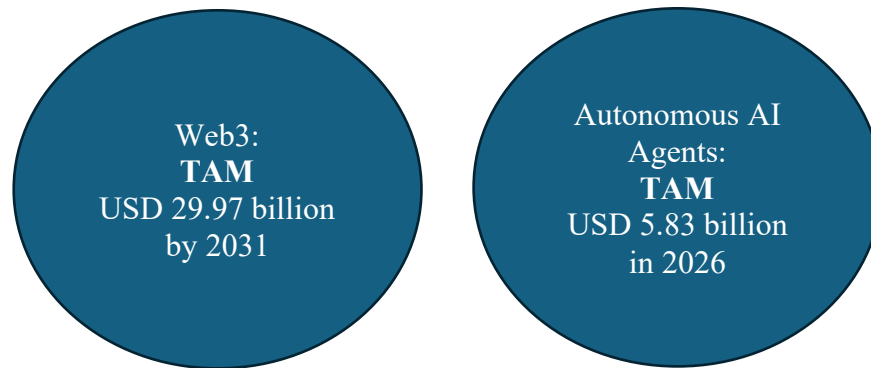


Figure 10: Total Addressable Markets of Web3 and Autonomous AI Agents

The ERC-4337 Account Abstraction Ecosystem grew in 2023 with 40 million smart accounts adopted across Ethereum and Layer 2 networks (Alchemy, 2026). The Web3 market is expected to grow from USD 4.97 billion in 2026 to USD 29.97 billion by 2031 (Mordor Intelligence, 2026).

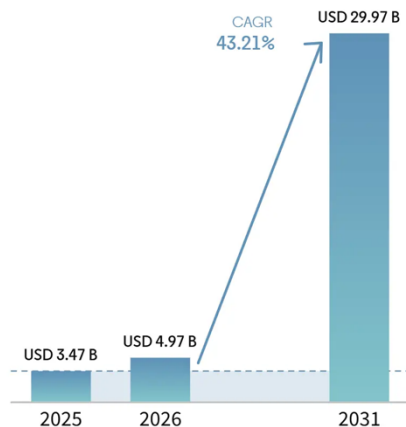


Figure 11: Web3 Market Size USD Billion (Mordor Intelligence 2026)

Composability has become a powerful tool for connecting blockchain protocols, smart contracts, and decentralized applications (dApps) within single-chain environments. Developers benefit from multiple protocols within single-chain environments like Ethereum, which enable the creation of sophisticated, user-friendly financial products (Okhaifo, 2024). However, it poses risks of slow speeds and high transaction (gas) costs, as blockchain systems are more complex, making them difficult for users to manage and understand, and less user-friendly due to their fragmented, rigid frameworks (Appendix F)

Biconomy's Market Sizing

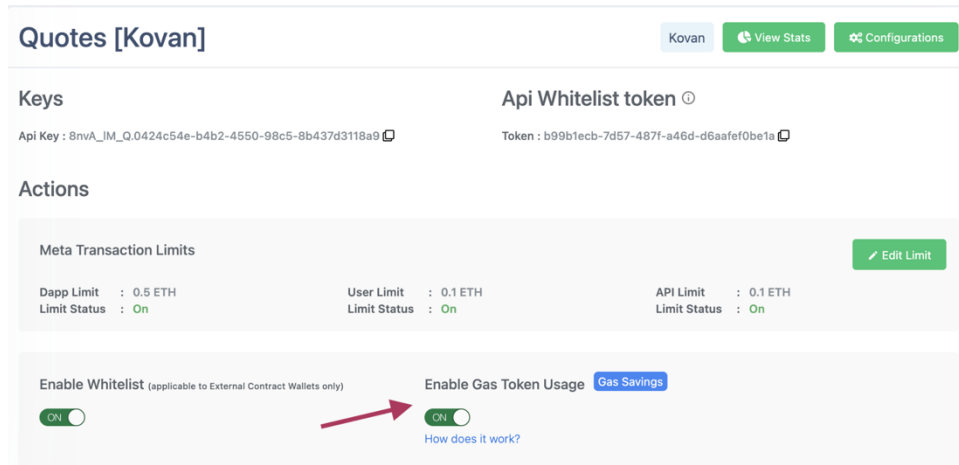


Figure 14: Biconomy- Simplifying Web3 Experiences (Defiprime, 2021)

Biconomy's Strategic Differentiator and Comparable Companies

Biconomy abstracts away blockchain complexity, enabling both humans and autonomous AI agents to interact with decentralized systems more seamlessly; this means users don't need to manage gas, and apps can automate execution. The Biconomy infrastructure naturally aligns with autonomous systems, machine-to-machine commerce, and programmable financial agents. Compared with competitors such as ZeroDev or Pimlico, which focus on infrastructure scale, institutional wallet security, or exchange ecosystems, Biconomy's focus is on transaction abstraction, programmable smart account infrastructure, and user-centric automation designed to support seamless Web3 and AI-agent interactions (Appendix E)

Comparable Companies	What they do well
 Alchemy	Expanding developer ecosystem, dependable infrastructure, enterprise scale (Alchemy, 2026).
 Safe	Reliable smart account security, multisig dominance (Safe Wallets, 2026).
 Coinbase	Large distribution with regulatory credibility (Coinbase, 2026)
 ZeroDev	Robust developer-first ERC-4337 tooling (ZeroDev, 2026).
 Pimlico	Offers a specialized version of ERC-4337 within its blockchain stack (Pimlico, 2026)
 Stackup	Developer-friendly account abstraction APIs (Stackup, 2026).
 Thirdweb	Simplified Web3 onboarding and SDK ecosystem (Thirdweb, 2026).

Figure 15: Competitors Comparable Analysis

Blockchain Infrastructure and Autonomous AI-Agent Adoption Industry Analysis

Biconomy operates in a highly competitive market where account abstraction infrastructure could potentially become foundational to autonomous, AI-driven commerce. Although the barriers to entry are moderate, the rollout massively depends on ecosystem adoption, developer trust, scalability, and the ability to stand out from larger infrastructure competitors such as Alchemy, Safe, and the Coinbase Developer Platform. The diagram below highlights the industry's attractiveness for Biconomy (Appendix E)

Business Model

Biconomy has been operating as a B2B2C infrastructure provider. Their customers are decentralized application (dApp) teams, Web3 developers, protocols, and traditional financial institutions that are testing on-chain settlements. Aave, Curve, dYdX, PancakeSwap, The Sandbox, Trust Wallet, Gemini Wallet, etc., are among their customers (Biconomy, 2026a; CoinMarketCap,

2026). End-users are not interacting with Biconomy directly; they are rather indirect beneficiaries. This makes Biconomy closer to Stripe or Twilio than to a consumer fintech app.

Revenue Generation

Biconomy generates revenue through four primary channels, all based on transaction fees. Biconomy earns a margin on each sponsored transaction on its Paymaster service, enabling decentralized applications (dApps) to cover users' gas fees or allowing users to pay gas fees in stablecoins rather than ETH (Biconomy, 2026b). The second is Bundler, which packages together before they hit the chain, and Biconomy runs that packaging infrastructure as a usage-priced service through its SDK and API (Biconomy, 2023; Biconomy, 2026b). Third, SDK and dashboard access, which developers plug in via API keys from the Biconomy dashboard, with pricing tied to transaction volume and feature tier.

The newest stream is the Modular Execution Environment (MEE), Biconomy's decentralized layer for "Super transactions", which are single-signature flows that span multiple chains and steps (bridge, then swap, then deposit in one click), on which Biconomy earns transaction fees (Biconomy, 2025a; Biconomy, 2026e). Node operators on the network charge fees in BICO for orchestrating these. An AI agent rebalancing a yield strategy across three chains pays the same Paymaster, Bundler, and MEE fees as a person would.

BICO token utility:

BICO serves as a coordination asset with four functions: network fees, staking and delegation, governance, collateral, and incentive alignment (Khanduri, 2021). Network fees are paid by node operators on the Biconomy chain itself. As a staking and delegation function, token holders delegate BICO to node operators in exchange for a share of rewards Khanduri, 2021; CoinMarketCap, 2026). As a governance feature, token holders vote on the protocol's parameters. Lastly, the collateral and incentive alignment function serves executors, validators, and liquidity providers in multi-sided networks. Developers are not obliged to hold the token to use Biconomy services, which separates BICO from a “pay-to-play” model and makes its valuation more defensible.

Metric	Value	Notes
Total Supply	1,000,000,000 BICO	Max supply fixed at 1B
Circulating Supply (Jun 2026)	1,000,000,000 BICO	All circulated

Current Price (Jun 2026)	\$0.026–\$0.046	Range across major exchanges
Market Capitalization	\$23-\$59 M	At current price × circulating supply
Fully Diluted Valuation	\$23 M	FDV = market cap
All-Time High Price	\$21.45 (Dec 2, 2021)	99.9 % below ATH
All-Time Low Price	\$0.018 (Dec 2025)	Dec 19, 2025
ICO Price	\$0.30	CoinList public sale, Nov 2021
Token Type	ERC-20 (Utility + Gov)	Governance voting & staking
Unlock Status	Fully Unlocked	Vesting schedule ended 2025

Figure 15: BICO Token Overview

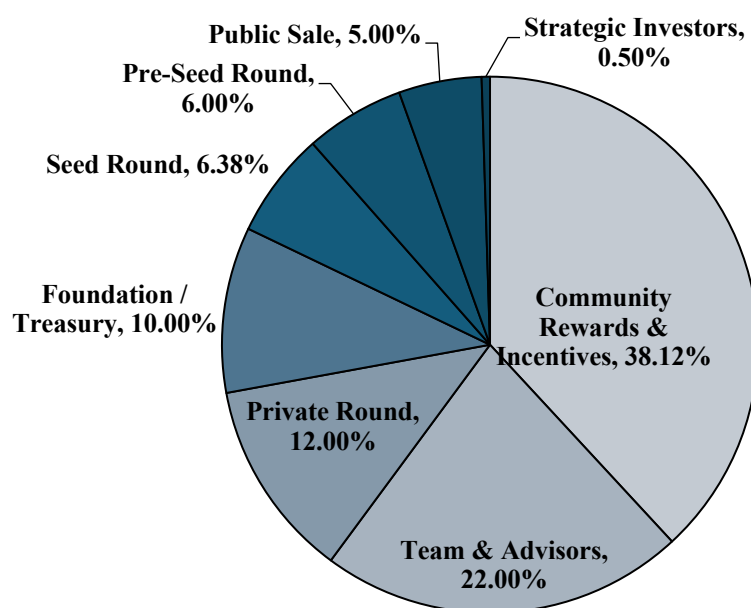


Figure 16: Bico Token Allocation by Category

Value creation

For users and agents, transactions start to feel like Web2. For AI agents specifically, smart accounts can be programmed with spending limits, conditional rules, and scoped permissions, which allow a user to trust an agent with their money (Biconomy, 2026d). Biconomy's ERC-8211 standard, proposed by the Ethereum Foundation in April 2026, takes this further. The MEE network compounds it; more nodes attract more dApps, more dApps drive more transactions, and more transactions reward more nodes. As revenue scales with on-chain activity rather than sales volume, the unit economics look more like those of a payment processor than those of a

SaaS company. A fee-per-transaction model benefits greatly as ecosystems grow and more users are drawn to the platform; the benefits are far larger than in traditional SaaS businesses.

Financial traction

Biconomy has raised \$10.5M across seed, Series A, and Series B stages (The Block, 2021; CB Insights, 2026; Tracxn, 2025). Series A in July 2021 of \$9M size led by Mechanism Capital and DACM with participation from Bain Capital, Coinbase Ventures, and others (The Block, 2021). The Series B round was completed in 2024, with participation from Jump Capital and Borderless Capital (Tracxn, 2025). The traction, as reported, includes more than 4 million end users onboarded through the SDK, over 400 integrated dApps, more than 17 million processed transactions, and over US\$1.4 billion in historical transaction volume (Biconomy, 2026a). The company has expanded integrations with Unichain, Plasma, and the XDC Network through 2025-26 (CoinMarketCap, 2025).

Metric	Value	Context
Unique Users Onboarded	> 5 million	As of 2024 via AA-SDK (Decrypt, Mar 2024)
Smart Accounts Created	> 1.5 million	Via Biconomy infrastructure (Decrypt, Mar 2024)
UserOps Processed	> 6.1 million	ERC-4337 UserOperations (Decrypt, Mar 2024)
dApps Integrated	> 400	Across 20+ EVM chains (Decrypt, Mar 2024)
Chains Supported	20+ EVM chains	Including Scroll, Blast, ZetaChain
AA Market Share (est., 2023)	8–50 %	Paymaster vol: 8%; UOps share 50% (BlockEden)
System Uptime Claim	99.9 %	Enterprise-grade SLA (CoinMarketCap)
Gas Tokens Supported	> 10,000 ERC-20	Users can pay gas in any ERC-20 token

Figure 17: Protocol Usage & Adoption

Team Analysis

Biconomy was founded in 2019 by Ahmed Al-Balaghi, Aniket Jindal, and Sachin Tomar (Appendix D). Sachin Tomar’s engineering background and product stack span ERC-4337, smart accounts, paymasters, bundlers, DAN, and MEE, demonstrating technical depth and the ability to build deep infrastructure for Web3 transaction processing (LinkedIn, 2026). Advisors such as Matthew Liu and Sandeep Nailwal, indicating broader operating structure, external credibility, and network effects, with execution evidence (LinkedIn, 2026). Although the business remains highly founder-led, with key-person dependency and moderate risk from concentration within the original core team.

Implementation Roadmap

If the ERC-8211 standard becomes the default way AI agents execute DeFi strategies, Biconomy benefits as the originator with the most mature production deployment in the market. EIP-8141 would be the bigger shift, as every Ethereum account would get smart account capabilities built in, and Biconomy would become part of the baseline. Nexus ecosystem expansion could lead to future distribution expansion. Each new wallet built on Nexus after Gemini raises switching costs and locks in network effects. MEE scaling marks a quieter enabler; more supported chains mean more user intents can be resolved, making every existing integration more useful.

An open question on their roadmap remains: the BICO token's utility. As of late May 2026, the token traded around \$0.023, below both its 7-day and 30-day moving averages, with further bearish technical momentum (CoinMarketCap, 2026). This weakness is primarily attributed to broad altcoin rotation and Bitcoin ETF outflows, rather than anything Biconomy-specific, meaning that the protocol's growing deployment numbers have not been enough to insulate the token from sector-wide pressure (Ibid).

This raises the question of whether the team can build a clearer link between protocol usage and token value. Based on current signals, Biconomy, as a platform and service, is poised to look very different in 2027 than it does today (Appendix G). The key question is whether the ERC-8211 becomes the default execution pattern for agent-driven DeFi. If ERC-8211 and EIP-8141 both ship and gain traction, Biconomy moves closer to a protocol-level standards body than just a startup, with revenue flowing through Nexus integrations and infrastructure fees rather than direct user transactions.

Risks, Compliance, and Mitigation Techniques for Biconomy AI

Regulation and Compliance

Biconomy's operations are at the intersection of crypto infrastructure, DeFi automation, and potentially AI-driven financial decision making (Biconomy, 2026). This creates regulatory uncertainty due to overlaps among multiple legal frameworks. These are: crypto-asset regulation, data protection law, financial promotion rules, consumer protection, and AI governance. In the EU, the Markets in Crypto-Assets Regulation (ESMA, 2026) has introduced harmonized rules for crypto-assets, covering transparency, disclosure, authorization, and supervision for crypto-asset

issuers. Raising concerns about Biconomy's infrastructure, which third parties use to facilitate payments or token transactions, the company itself may face indirect regulatory exposure. To mitigate this significant risk, Biconomy should maintain a "compliance-by-design" architecture (KPMG, 2022). This means separating their infrastructure provision from regulated investment advice. In turn, this would require partnerships with dApps to disclose their regulatory status, integrate KYC/AML protocols, and, most importantly, clearly state that AI agents support execution rather than guarantee investment outcomes.

Privacy and Data Security

AI and blockchain applications often rely on user data, transaction history, wallet behavior, and DeFi activity. This creates privacy concerns because blockchain data is publicly accessible and difficult to remove. For Biconomy, this creates potential exposure to GDPR or UK GDPR requirements, especially if automated decision-making by AI agents is involved. More specifically, Article 22 of the GDPR gives users protections against solely automated decisions (GDPR, 2017). The key concern is data leakage. Even if wallet addresses are pseudonymous, combining on-chain transaction data with off-chain behavioral data can reveal user identity or financial behavior. This risk could be reduced by Biconomy, which minimizes the collection of personal data and avoids storing sensitive user data on-chain. In terms of AI features, Biconomy should use off-chain encrypted processing, strict access controls, and privacy impact assessments. In addition, ZKPs should be used to verify user eligibility.

AI Bias and Model Reliability

As Biconomy uses and supports AI agents in DeFi, there is a risk that their AI models could make biased or inaccurate decisions. The EU AI Act aims to create a framework for trustworthy AI, and more specifically, high-risk AI systems are expected to meet risk management requirements, data governance, transparency, and human oversight (EUROFI, 2024). Compliance is particularly important given the volatility and rapid changes in DeFi markets. For example, if a model is trained on historical transaction patterns, it may fail during market stress or liquidity shocks. AI reliability requires continuous monitoring. To mitigate this risk, Biconomy should introduce AI model governance. This means implementing pre-launch testing, bias audits, stress testing, and performance monitoring. AI agents shouldn't be given full authority over users' funds; instead,

they should operate within user-predefined spending limits, stop-loss rules, and, most importantly, mandatory human oversight over high-risk transactions.

Adoption and User-Education

Furthermore, by analyzing Biconomy's value proposition, it becomes clear that the company aims to make Web3 more user-friendly by reducing friction, gas costs, and user onboarding challenges (Biconomy, 2026). However, concepts such as smart accounts and AI agents may remain difficult for inexperienced users to fully understand. If new users don't understand what an AI agent is authorized to do, they may approve poorly understood transactions. As blockchain interactions become simpler, it becomes easier for users to underestimate the risks of their actions. To mitigate, Biconomy should create example transaction previews to help users fully understand the implications of each transaction. This could reduce user error while improving transparency around automated transactions. For example, some previews when an AI agent executes a DeFi strategy could show the expected action, the DeFi protocol used, the estimated fees, and the cancellation options.

Operational Complexity and Scaling

Biconomy's infrastructure supports reliable transaction execution across multiple chains, dApps, and wallets (Biconomy, 2026). Multi-chain execution is beneficial for users, as it allows them to interact across different chains with fewer steps, but it also creates operational complexity. Key operational risks include downtime, congestion, and failed transaction execution. This is important for AI agents, as automated systems can execute large numbers of transactions at high speed. But if the infrastructure scales poorly, Biconomy may face higher failure rates and weaker user experience. To minimize these risks, it is crucial for Biconomy to build redundancy into the bundler and paymaster infrastructures and to apply rate limits to AI-agent execution. In addition, if automated execution fails or is slow, Biconomy should create fallback systems that allow users to manually execute transactions manually.

Smart Contracts and Technical Execution

Biconomy's platform relies on smart contracts and account abstraction technology to make blockchain transactions easier and more automated for users (Biconomy, 2026). Even a single mistake in the smart contract code can affect users' funds or transactions. As blockchain transactions are largely irreversible, coding vulnerabilities could expose users to permanent asset

loss or failed execution. For example, Biconomy uses technologies like smart accounts, bundlers, and paymasters to automate transactions and gas payments (Biconomy, 2026). In addition, because Biconomy supports multi-chain execution, the entire infrastructure becomes more complex, and mistakes could be more detrimental to users. Risk exposure could be reduced through independent smart contract audits, continuous monitoring systems, and staged deployment testing prior to protocol updates. Finally, as mentioned above, AI agents should also operate under strict rules, such as transaction caps and user approval requirements for risky transactions.

Conclusion

The next generation of fintech will be defined by intelligence and execution working together. The model is shifting from a human-centric approach to financial tools to a human-centric goal-setting approach, with autonomous systems managing financial activity for humans. In combination, its account abstraction layer, smart accounts, gasless transaction support, and ERC-4337 execution environment offer the programmable execution layer needed. Biconomy has created a programmable financial environment that supports delegated permissions, auto-execution, and cross-chain coordination.

This positioning is a strategic advantage, as incumbent institutions are unlikely to invest in this infrastructure. While traditional banks still use legacy systems to approve transactions, many crypto wallet providers are keen to enhance the user experience for human users, not for autonomous agents (Arshadi and Dombrowski, 2026). Consequently, the execution layer to support AI-driven finance will likely be developed by niche infrastructure companies that can connect the artificial intelligence and programmable blockchain system.

Although Biconomy offers a compelling solution, it is hindered by significant risks. Regulatory uncertainty, smart contract and infrastructure failures, and adoption friction remain key hurdles. Technical risks are omnipresent, as bugs, congestion, or chain complexity could disrupt execution. They operate in a competitive, rapidly evolving crypto market, where growth depends on rapid innovation.

The conclusion is clear: AI cannot deliver autonomous finance without trustless execution, and blockchain cannot deliver intelligent finance without AI. This analysis observes that Biconomy is a credible execution layer for autonomous finance, but its long-term success depends on the broad adoption of AI agents and sustained crypto usage.

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Appendix

Appendix A: Historical Timeline of Biconomy (Malak, 2021), (YourStory, 2019), (Kessler & Keoun, 2024), (Javor, 2025), (Javor, 2025), (Ranko, 2025), (Capozzalo, 2026)

Date	Milestone	Why it matters
2019-2021	Founding, gasless transactions	Founded in Dubai by Ahmed Al-balaghi, Aniket Jindal, and Sachin Tomar. The first product was meta-transactions and gasless UX for dApps. From day one the company sold to developers, not to end users
2021	BICO token launch	Native token for network fees, governance, and staking. The launch funded the smart-account stack that came next
2022-2023	Smart Account v2 on ERC-4337	Biconomy ships a full account-abstraction stack on ERC-4337. The company transforms from a transaction relay to a more holistic smart-account platform
Jun 2024	DAN launch	The Delegated Authorization Network goes live. The user's key is sharded across EigenLayer-secured nodes. This allows an AI agent to sign transactions without ever holding the raw private key. This marked an important step toward a more credible security model for on-chain agents.
Nov 2024	Klaster acquisition, MEE preview	Biconomy buys Klaster to build the Modular Execution Environment. A user intent can now be turned into a multi-step path across chains.
Dec 2024	Nomy intent agent	Biconomy's own AI agent launches. Users can state a goal in plain English; the AI agent breaks this into actions that run through DAN and MEE. The whole stack has now become visible end-to-end as a single product.
2025	Nexus on ERC-7579	Nexus replaces Smart Account v2 as the company's flagship product. It is built on the ERC-7579 modular standard, audited by Spearbit and Cyfrin, and scored 92% by DefiSafety.

Aug 2025	Gemini Wallet on Nexus	Gemini's self-custodial passkey wallet ships on the Nexus stack. This represented the first major US centralized exchange to run on Biconomy's infrastructure
Apr 2026	ERC-8211 published	The standard effectively puts a name on the smart-batching pattern Biconomy has been running in production for over a year. The company is now the named originator of the Ethereum standard.

Appendix B: Biconomy Infrastructure Supports Blockchain

Mexa Examples:

- Curve.Fi: for example, a decentralized Exchange like Uniswap uses Biconomy to support Meta transactions for gasless BTC deposits.
- Decentral Games: a play-to-earn game in the metaverse using Biconomy on Matic Network, providing gasless transactions.

Hyphen Mechanics:

- Hyphen leverages the Biconomy network of validators and executors. Each chain has a liquidity pool that earns a portion of the transfer fee for providing liquidity to cross-chain transfers. Periodic rebalancing is low when liquidity on a particular chain is low, to improve capital efficiency.

Forward Mechanics:

- Biconomy relayers handle the ETH gas fees in the backend. They pay the optimal gas fees for each transaction and ensure they are never stuck. If congestion occurs, relayers automatically detect and resolve it by gradually increasing gas fees. This improves the user experience, as users do not end up with transactions that are temporarily stuck or on hold.

Appendix C: Top-Down Market Sizing Logic

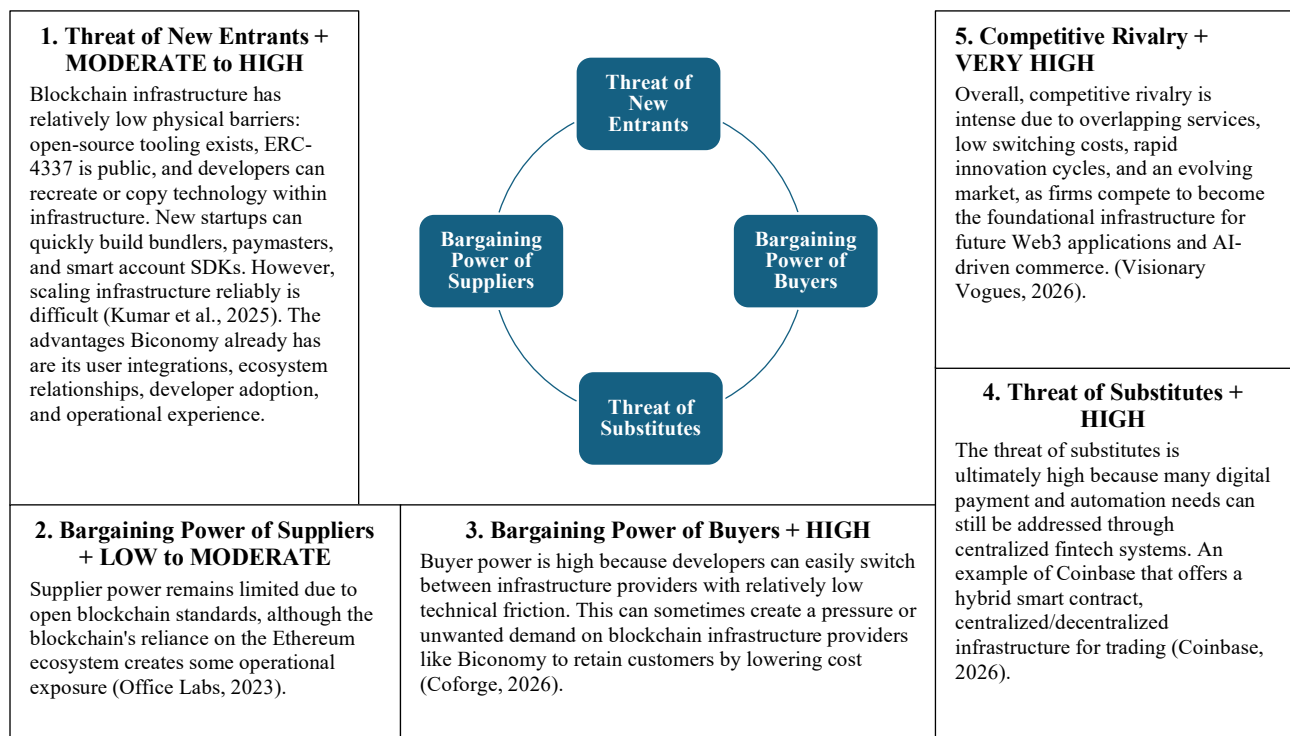
Metric	Calculation	Value (USD)
TAM	Web3 market size (2026)	4.97 bn
SAM	USD 4.97 bn $\times$ 62.4% (Adoption)	3.10 bn
SOM	3.10 bn $\times$ 8% (Market Capture)	24.81m

Appendix D: Team Analysis (Gompers et al, 2020 and Pitchbook, 2026)

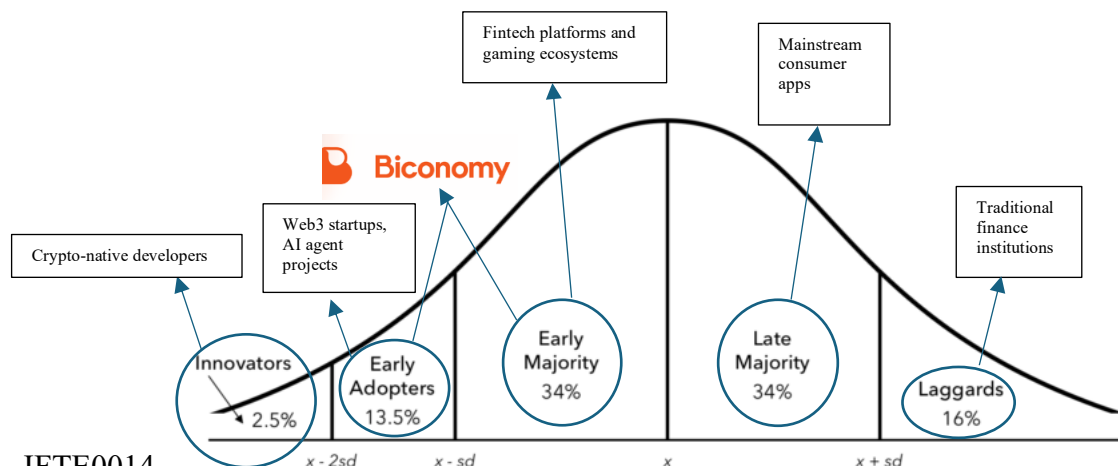
Team dimension	Evidence from Biconomy	What it signals	Assessment
Founder experience	Biconomy was founded in 2019 by Ahmed Al-Balaghi, Aniket Jindal, and Sachin Tomar. Ahmed is the CEO, Aniket is COO, and Sachin is CTO.	The founding team has stayed intact and has clear functional role separation.	Strong. Stable founding structure is a positive signal for execution and governance.
Technical depth	Sachin Tomar is listed as Co-Founder and CTO, and has an engineering background and product stack spans smart accounts, paymasters, bundlers, DAN, and MEE.	The team appears capable of building deep infrastructure, not just a wrapper product.	Strong. The technology is complex, so technical leadership matters materially.
Domain relevance	The company focuses on Web3 transaction infrastructure, account abstraction, and AI-agent execution. The founding team is consistently presented as blockchain-native and web3-focused.	The team is operating inside a domain where product-market fit depends on niche technical knowledge.	Strong. This is a domain where generalist product teams are weaker.
Complementary roles	The top team separates CEO, COO, CTO, and people leadership, with advisors including Matthew Liu and Sandeep Nailwal.	Indicates a broader operating structure beyond the founders alone.	Positive. The mix suggests early institutionalization and external credibility.
Execution evidence	Biconomy claims more than 4 million end users, 400+ integrated dApps, 17 million+ processed transactions, and over US\$1.4 billion in historical transaction volume.	Strong proof that the team has moved beyond concept into deployment.	Strong. Traction reduces “idea-only” risk.
Advisor quality	Advisors listed include Matthew Liu and Sandeep Nailwal. Matthew Liu is a known crypto founder, and Sandeep Nailwal is a major Web3 figure.	High-quality advisors can improve network access, credibility, and strategic guidance.	Positive. Particularly useful in crypto, where trust and network effects matter.
Team scale and maturity	Public directories show a broad team across product, growth, engineering, partnerships, and people functions.	The company appears to have moved past a three-founder startup into	Positive. That supports scalability and go-to-market execution.

		a more mature operating team.	
Risk / concentration	The business still appears highly founder-led, and the most strategically important roles remain concentrated around the original core team.	Key-person dependency may still be meaningful.	Moderate risk. Strong founders, but execution may still be concentrated.

Appendix E: Porter's 5 Forces



Appendix F: Diffusion of Innovation Adoption Curve (New York: Free Press, 2003)



Appendix G: Next Steps Timeline

ERC-8211 Adoption	• If ERC-8211 becomes the default way agents execute DeFi strategies, Biconomy benefits as the originator with the most mature implementation • The Ethereum Foundation is actively backing the standard
EIP-8141 pairing	• EIP-8141 is a new Ethereum transaction type that bakes smart-account features into the protocol itself. • If both ship, Biconomy stopped being an opt-in stack and becomes a baseline capability.
Nexus Ecosystems	• Each new integration depends on the network effect of the ecosystem. • It also makes a migration to a competing stack more expensive. Distribution is the moat.
MEE Scaling	• More chains supported means more user intents that can resolve • This strategy gives NOME reach and makes every existing integration more valuable.
BICO token utility	• BICO currently is down roughly 46% into early 2026 even though smart account deployments accelerated over the same period. • A tokenomics refresh is one of the most plausible 2026 events.